

SUMMARY OF PRODUCT CHARACTERISTICS

1. NAME OF MEDICINAL PRODUCT

Tricef 20 mg/mL powder for oral suspension.

2. QUALITATIVE AND QUANTITATIVE COMPOSITION

Each 5 mL of oral suspension contains 100 mg of cefixime.

Excipient with known effect:

Sucrose – 505.022 mg/mL

For the full list of excipients, see section 6.1

3. PHARMACEUTICAL FORM

Powder for oral suspension.

Powder for oral suspension, white to pale cream colored and strawberry odor.

4. CLINICAL PARTICULARS

4.1 Therapeutic indications

Tricef is indicated for the treatment of the following infections caused by susceptible agents (see section 4.4 and 5.1):

- Acute otitis media
- Upper respiratory tract infections (pharyngitis, tonsillitis, acute sinusitis)
- Lower respiratory tract infections (acute exacerbations of chronic bronchitis, community acquired pneumonia);
- Urinary tract infections (acute cystitis, uncomplicated acute pyelonephritis);
- Urethritis and uncomplicated gonococcal cervicitis.

Consideration should be given to national and/or local guidelines on the appropriate use of antibiotics.

4.2 Posology and method of administration

The usual recommended posology is 8 mg/kg, single dose.

In terms of the pharmaceutical form, the dosage is generally as follows:

- *Children aged 2-4 years*: 1 measuring spoon every 24h; ½ measuring spoon every 12 hours.
- *Children aged 5-8 years*: 2 measuring spoons every 24h; 1 measuring spoon every 12 hours.
- *Children aged 9-12 years*: 3 measuring spoons every 24h; 1½ measuring spoon every 12 hours.

- *Patients with renal impairment:* The medicine may be administered to patients with impaired renal function. Doses indicated above may be given in patients with creatinine clearance of 20 mL/min or above. In patients whose creatinine clearance is less than 20 mL/min, it is recommended not to exceed a daily dose of 200 mg. This dose should also not be exceeded in patients undergoing chronic peritoneal dialysis or hemodialysis, since cefixime is slowly removed from circulation by dialysis.

In patients with impaired liver function, there is an increase on time required for reaching the maximum serum concentration of cefixime, however this does not require changes of posology in these patients.

Concomitant administration with food may increase the time required to reach the C_{max} not interfering with AUC₂₄ or T_{1/2β}, effect with no clinical relevance.

4.3 Contraindications

Hypersensitivity to the active substance and, in general, to beta-lactam antibiotics or to any of the excipients mentioned in section 6.1.

Tricef contains 2.52 g of sucrose per 5 mL of oral suspension. Patients with hereditary fructose intolerance, glucose-galactose malabsorption syndrome or sucrase-isomaltase insufficiency should not take this medicine.

4.4 Special warnings and precautions for use

Tricef should be used with caution during pregnancy and lactation, the contraindications related to hypersensitivity to cefixime or its excipients should be respected, and also the dose reduction in renal impaired patients should be observed.

Cephalosporins should be given with caution to penicillin-sensitive patients, as there is some evidence of cross-allergy between penicillins and cephalosporins, as severe reactions (including anaphylaxis) to both classes have been observed (see section 4.3).

Cases of severe skin reactions with cefixime, as toxic epidermal necrolysis, Stevens-Johnson syndrome or rash with eosinophilia and systemic symptoms (DRESS) were reported. If a severe adverse skin reaction occurs, the use of cefixime should be immediately discontinued and appropriate therapeutic measures should be triggered.

Like other cephalosporins, cefixime may also lead to acute renal impairment, including interstitial nephritis. When acute renal impairment occurs, cefixime should be discontinued and appropriate therapeutic measures should be adopted.

Tricef should be used with caution particularly in the presence of severe renal impairment (see section 5.2).

Prolonged use of cefixime may cause overgrowth of non-sensitive agents. The treatment with broad spectrum antibiotics alters the normal flora of the colon and can lead to the colonization by *Clostridium* strains. Studies indicate that the toxin produced by *Clostridium difficile* is the major cause of antibiotic associated diarrhea. The pseudomembranous colitis is associated with the use of broad spectrum antibiotics (macrolides, semi-synthetic

penicillins, lincosamides and cephalosporins including cefixime). It is important to consider this diagnosis in patients who develop diarrhea associated with antibiotic use.

Some patients with severe diarrhea due to pseudomembranous colitis, developed during or after the use of cefixime, have been at risk of life, thus it should be considered (see section 4.8). In case of suspicion, the use of cefixime should be discontinued and appropriate treatment measures should be initiated. Digestive endoscopic procedures, such as sigmoidoscopy or bacteriological might be necessary. Treatment measures include fluids, electrolytes and protein supplements. If colitis does not ameliorate after drug discontinuation or if symptoms get worse, treatment with oral vancomycin is indicated. This is the antibiotic of choice in pseudomembranous colitis by *C. difficile*. Other causes of colitis should be excluded. The use of drugs that inhibit intestinal peristalsis is contraindicated.

4.5 Interaction with other medicinal products and other forms of interaction

Antacids do not interfere with the absorption of cefixime. The tubular reabsorption inhibitors such as probenecid may hinder the urinary excretion of cefixime by increasing the values of C_{max} and AUC₂₄.

The salicylates and other nonsteroidal anti-inflammatories can displace cefixime from its plasma protein binding, thereby increasing the concentration of free fraction.

In common with most cephalosporins, increases in prothrombin times have been noted in a few patients. Thus, caution is recommended in the administration of cefixime in patients undergoing anticoagulant treatment and adjustment of frequency of monitoring of the International Normalized Ratio (INR).

The administration of cefixime may reduce the effectiveness of oral contraceptives. It is therefore recommended to take additional non-hormonal contraceptive measures.

False positive results can be observed in the urine glucose determination with cupric reagents, but not by those using glycooxidase. A false positive Coombs test may also be presented as it happens with most of cephalosporins.

4.6 Fertility, pregnancy and lactation

In preclinical trials no differences were detected between the group of animals of control and the group that received the drug, with respect to fertility parameters, namely mating behavior, pregnancy rate, duration of pregnancy or delivery. The transplacental transfer of cefixime was about 1% of the dose administered to pregnant rats. The transfer through breast milk was about 1.5% of the total given to the mother.

Although animal experience does not suggest any kind of toxicity during pregnancy, the harmlessness of cefixime during pregnancy in humans is unclear. Tricef should not be used during pregnancy and lactation unless the doctor considers its use essential.

4.7 Effects on ability to drive and use machines

No effects were observed on the ability to drive and use machines.

4.8 Undesirable effects

The undesirable effects (according to the results of clinical trials) are listed in order of decreasing seriousness within each frequency class.

The frequency, type and severity of adverse reactions in children are expected to be the same as in adults.

In this section the following convention was used for the classification of undesirable effects in terms of frequency:

- Very common ($\geq 1/10$)
- Common: ($\geq 1/100$, $<1/10$)
- Uncommon: ($\geq 1/1,000$, $<1/100$)
- Rare: ($\geq 1/10,000$, $<1/1,000$)
- Very rare: $<1/10,000$
- Not known (cannot be estimated from the available data)

MedDRA Class	System	Organ	Adverse Reaction	Frequency
Infections and infestations			Bacterial superinfection	Rare
			Fungal superinfection	Very rare
			Antibiotic-associated colitis (see section 4.4)	
Blood and lymphatic system disorders			Eosinophilia	Rare
Immune system disorders			Hypersensitivity	Rare
			Anaphylactic shock, rheumatoid arthritis	Very rare
Metabolism and nutrition disorders			Anorexia	Rare
Nervous system disorders			Headache	Uncommon
			Vertigo	Rare
			Psychomotor hyperactivity	Very rare
Gastrointestinal disorders			Diarrhea	Common
			Abdominal pain, nausea, vomiting	Uncommon
			Flatulence	Rare
Hepatobiliary disorders			Hepatitis, jaundice	Rare
Skin and subcutaneous disorders			Rash	Uncommon
			Rash with eosinophilia and systemic symptoms (DRESS)	Rare
			Erythema multiforme	Very rare
			Pruritus	
			Stevens-Johnson Syndrome	
			Toxic epidermal necrolysis	
			Hives	
Renal and urinary disorders			Interstitial nephritis	Very rare
General disorders and administration site conditions			Mucosal inflammation, fever	Rare

Investigations	Hepatic enzyme increased (transaminase, alkaline phosphatase)	Uncommon
	Bloodurea increased	Rare
	Blood creatinine increased	Very rare

4.9 Overdose

There is no information about cases of overdose. Doses of 2 g in healthy adults caused moderate gastrointestinal effects as those seen with therapeutic doses.

In case of acute poisoning it is likely to occur diarrhea, vomiting and abdominal pain.

No specific antidote exists. Gastric lavage is indicated (if intake occurred for less than 2 hours) and the patient should be hydrated and, if necessary, electrolytically balanced.

Peritoneal dialysis and hemodialysis do not remove the antibiotic from the circulation in clinical significant quantities.

5. PHARMACOLOGICAL PROPERTIES

5.1 Pharmacodynamic properties

Pharmacotherapeutic group: 1.1.2.3 - Anti-infectious drugs. Antibacterials. Cephalosporins. 3rd generation cephalosporins, ATC code: J01D D08 cefixime

Mechanism of action

Cefixime, the only active substance of Tricef, is a bacterial agent belonging to the 3rd generation cephalosporins class.

As a beta-lactam antibiotic, acts by inhibiting bacterial cell wall synthesis. It is therefore a bactericidal antibiotic. Due to the introduction of a carboxymethoxy-imino radical in position 7 of cephemic core, cefixime has a high resistance to inactivation by most of the beta-lactamases produced by gram-positive or gram-negative bacteria. This characteristic translates into a similar activity on susceptible bacteria, whether they are or not beta-lactamases producers.

PK/PD Relationship

The time that the plasma concentration of cefixime exceeds the MIC (minimum inhibitory concentration) of the infecting organism has

been shown to best correlate with efficacy in PK/PD studies.

The best therapeutic response is estimated to occur when T> MIC is at least 40-50% of the interval between doses.

Mechanism of resistance

Bacterial resistance to cefixime may be due to one or more of the following mechanisms:

- Hydrolysis by beta-lactamases, induced or de-repressed in certain aerobic gram-negative bacterial species;
- Reduced affinity of penicillin binding proteins for cefixime;
- Reduction in permeability of the outer membrane of Gram-negative bacteria;
- Expression of membrane drug efflux pumps.

More than one of these mechanisms of resistance may co-exist in a single bacterial cell.

Depending on the mechanism(s) present, bacteria may express cross-resistance to several or all beta-lactam antibiotics or to antibiotics of other classes.

Values of critical concentrations (breakpoints)

The breakpoints established for cefixime by EUCAST (European Committee on Antimicrobial Susceptibility Testing) in April 2010, are:

- *H. influenzae*: sensitive ≤ 0.12 mg/L, resistant > 0.12 mg/L
- *M. catarrhalis*: sensitive ≤ 0.5 mg/L, resistant > 1.0 mg/L
- *Neisseria gonorrhoeae*: sensitive ≤ 0.12 mg/L, resistant > 0.12 mg/L
- *Enterobacteriaceae*: sensitive ≤ 1.0 mg/L, resistant > 1.0 mg/L (for uncomplicated urinary tract infections only). Screening of the production of ESBL (extended spectrum beta-lactamases) may be necessary for infection control, surveillance or epidemiology purposes.

The prevalence of acquired resistance may vary geographically and with time. For selected species, local information on resistance is desirable particularly when treating severe infections. Expert advice should be sought when the local prevalence is such that the efficacy of the agent, in some types of infections, is questionable.

Susceptible species
Aerobes, Gram positive <i>Streptococcus pneumoniae</i> (penicillin-susceptible) <i>Streptococcus pyogenes</i>
Aerobes, Gram negative <i>Escherichia coli</i> % <i>Haemophilus influenzae</i> <i>Klebsiella species</i> % <i>Moraxella catarrhalis</i> <i>Proteus mirabilis</i> %
Species for which acquired resistance may be a problem <i>Enterobacter species</i>
Naturally resistant species <i>Clostridium difficile</i> <i>Bacteroides fragilis</i> <i>Enterococci</i> <i>Pseudomonas species</i> <i>Staphylococcus aureus</i> + <i>Streptococcus pneumoniae</i> (penicillin-resistant)

% ESBL producing isolates are always resistant

+Cefixime has poor activity against staphylococci (regardless of susceptibility to methicillin)

5.2 Pharmacokinetic properties

Absorption

The presence of a vinyl radical in position 3 gives cefixime a satisfactory ability to be orally absorbed. In fact, the oral bioavailability of this antibiotic is about 48-50%. The presence of food does not affect the oral absorption of cefixime. T_{max} is slightly wider, however the C_{max} and AUC₂₄ are not modified.

Repeated therapeutic doses do not cause accumulation in the body, either in adults or in children.

Distribution

Serum protein binding of cefixime is about 70% and the free fraction diffuses well to various body tissues, except CNS. The antibiotic penetrates well into the maxillary sinuses, middle ear, respiratory tract (including the bronchial secretions), tonsils, prostatic fluid and tissue and other organs.

After a dose of 400 mg, in adults, the concentrations in the tissue of the gallbladder and the bile are approximately 20 and 190 mcg/mL, respectively, 4 to 12 hours after ingestion of the antibiotic. Urinary concentrations are also high.

Biotransformation

Cefixime undergoes moderate metabolization in the body, and is essentially excreted, in active form, mainly by the bile, and in appreciable amount (20 to 30% of the absorbed dose) in urine.

The excretion of cefixime is slow, whereby the T_{1/2β} is quite long (about 4 hours), which ensures a good coverage compared to susceptible bacteria in the 24 hours following a single administration of 400 mg in adults or 8 mg/kg in children.

Elimination

Concentrations in urine and bile are very high, since the elimination takes place through these pathways.

Special Populations

- Children:

There is a longer half-life, a higher AUC₂₄ and an increased urinary excretion in infants when compared with older children, whose pharmacokinetic parameters are comparable to those of adults.

- Elderly patients;

In elderly patients (65-74 years), C_{max} and AUC₂₄ have values slightly higher than in young adults (20 to 32 years). However, these differences do not justify a lower dose for the elderly.

- Hepatic impairment:

No change was found in C_{max} or AUC₂₄ in cirrhotic patients after administration of 200 mg of cefixime, however the time for C_{max} and T_{1/2β} are increased, as well as renal clearance. No metabolites were detected in serum or urine.

- Renal impairment:

Although only 20 to 30% of the absorbed dose is excreted in the urine, C_{max} and $T_{1/2\beta}$ increase, for a given dose, as renal impairment is more severe. Thus, if these increases are not enough to require a dose reduction, for a creatinine clearance greater than 20 mL/min, the same does not happen when creatinine clearance is 20 mL/min or lower. In these latter circumstances C_{max} can double and $T_{1/2\beta}$ almost tripling and it is then necessary to reduce the dosage by half (usually to 200 mg/day for adults and 4 mg/kg daily for children). Hemodialysis and peritoneal dialysis removes from blood only small amounts of cefixime, a fact that has no clinical relevance.

5.3 Preclinical safety data

The studies conducted in rodents and dogs have shown that cefixime is devoid of toxic effects at single or repeated administrations. There were not detected mutagenic or clastogenic properties. The capacity and reproductive behavior of the tested animals was not altered. In rats and mice at cefixime is not teratogenic. The long-term carcinogenic effect was not assessed.

6. PHARMACEUTICAL PARTICULARS

6.1 List of excipients

Sucrose
Xanthan gum
Sodium benzoate
Strawberry flavor

6.2 Incompatibilities

Not known.

6.3 Shelf life

Shelf life: 2 years
Shelf life after reconstitution: 14 days.

6.4 Special precautions for storage

Store at temperature below 25°C.
Storage conditions of the reconstituted medicinal product, see section 6.3.

6.5 Nature and contents of container

Cardboard containing an amber type III glass bottle, a measuring spoon (volume of 5 mL) and a leaflet. Bottles with 60 mL.

6.6 Special precautions for disposal and other handling

Reconstitution: Shake the bottle to loosen powder . Add 4 measuring spoons of water. Stopper and shake vigorously. Fill with water up to the arrow ► marked in the label. Shake it until a homogeneous suspension is obtained.

No special requirements for disposal.

Any unused medicinal product or waste material should be disposed of in accordance with local requirements.

7. MARKETING AUTHORISATION HOLDER

Bialport - Produtos Farmacêuticos, S.A.
À Av. da Siderurgia Nacional
4745-457 S. Mamede do Coronado
Portugal

8. MARKETING AUTHORISATION NUMBER(S)

Registration N.: 9725200 – 60 mL; powder for oral suspension; 20 mg/mL; amber type III glass bottle

9. DATE OF FIRST AUTHORISATION/RENEWAL OF THE AUTHORISATION

Date of first authorization: 06th October 1989
Date of last renewal: 18th December 2012

10. DATE OF REVISION OF THE TEXT